

ZEYU LING

AI Algorithm Researcher

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WORK EXPERIENCE

Tencent - Hunyuan Lab - 3D Generation Center

Algorithm Researcher (Qingyun Program)

2025.12 – Present Shenzhen, Guangdong

- Lead the HYMotion M2M project, developing a unified foundation model for motion repair, keyframe in-betweening, and precise controllable motion generation

ByteDance - V-AI

Algorithm Researcher (Intern)

2025.6 – 2025.12 Hangzhou, Zhejiang

- Talking Face video generation based on WAN2.2-TI2V-5B
- Lip Sync based on WanVace (audio-driven video lip editing)
- Multi-view texture generation for 3D models based on WanVace

Zhejiang Lab

Algorithm Researcher (Intern)

2024.1 – 2024.12 Hangzhou, Zhejiang

- Led a large-scale video generation model project, designing and implementing Text2Video solutions with Diffusion+UNet and Diffusion+Transformer architectures

YITU Technology

Algorithm Researcher (Intern)

2023.3 – 2023.6 Hangzhou, Zhejiang

- Deployed YITU's CLIP-based vision foundation model in security scenarios
- Participated in optimizing multimodal retrieval / text-to-image search pipelines, improving business-side recall and ranking

ACADEMIC PROFILE

PRISM: Streaming Human Motion Generation with Per-Joint Latent Decomposition

First Author - arxiv.org/abs/2603.08590

Arxiv

- Proposed a joint-factorized 2D latent space (causal VAE + FK supervision), aligning the latent structure with human joints and significantly improving motion generation quality
- Through per-token timestep noise-free condition injection and self-forcing training, a single model unifies text-to-motion, pose-conditioned, and autoregressive streaming generation; training on clips of ≤ 360 frames can stably generate 10+ segments of long motion, achieving comprehensive SOTA

VersatileMotion: A Unified Framework for Motion Synthesis and Comprehension

First Author - arxiv.org/abs/2411.17335

ECCV2026

- Proposed VerMoVQ: a 2D causal spatio-temporal body-token VQ-VAE that preserves multi-person spatial information in world space and removes the bottleneck for extending the discrete-AR paradigm to multimodal and multi-person scenarios
- The first unified motion model competitive on 9+ tasks (single-/two-person, text/audio/motion, generation+understanding), revealing the compositional advantages of VQ+AR for unified motion
- Built MotionHub (~400K clips, multimodal annotations, unified SMPL) and the MotionBench multi-task benchmark

MCM: Multi-condition Motion Synthesis Framework for Multi-scenario

First Author - arxiv.org/abs/2404.12886

IJCAI2025

- Transformer Diffusion Model based on multi-axis self-attention, achieving SOTA on HumanML3D and KIT
- Supports injecting audio modality information into any text-to-motion pretrained Diffusion Model, enabling efficient transfer to Music-to-Dance, Speech-to-Gesture, and related tasks

SyncLipMAE: Contrastive Masked Pretraining for Audio-Visual Talking-Face Representations

First Author - arxiv.org/abs/2510.10069

Arxiv

EPIC: Efficient Prompt Interaction for Text-Image Classification

Second Author - arxiv.org/abs/2507.07415

ICME2025 - Oral

- Proposed EPIC, enabling efficient multimodal alignment through temporal prompts and similarity interaction while training only 1% of the base model parameters
- Achieved SOTA on UPMC-Food101 / SNLI-VE and comparable results on MM-IMDB, providing an efficient fine-tuning approach for downstream large-model tasks

EnchantDance: Unveiling the Potential of Music-Driven Dance Movement

Third Author - arxiv.org/abs/2312.15946

TMM 2026

- Built a strong dance latent space + dance diffusion model, addressing temporal redundancy in dance generation and significantly improving quality, diversity, and style consistency
- Released the ChoreoSpectrum3D dataset (70.32 hours, 4 dance styles), and enhanced music-dance consistency through a music style prediction network

Physics-grounded diffusion translates multi-granular specifications into metamaterial designs with high fidelity

Third Author - github.com/Leader-sheng/HyperDiffusionMeta

Nature-ML (under review)

- Pioneered a two-stage framework, "PDE-constrained physical-field generation $\rightarrow$ hypergraph structure generation," enabling cross-modal design from multi-granular text to physical fields and then to structures
- Uses hypergraphs to model higher-order interactions, while physical field-structure coupling ensures physical consistency and drives innovation in aerospace, biomedicine, and related fields

EDUCATION

Ph.D. - Computer Science and Technology

Zhejiang University - College of Computer Science and Technology - CAD&CG

2023.12 – Present

- 2024 Five-Good Graduate Student, Outstanding Graduate Student

M.S. - Software Engineering

Zhejiang University - School of Software Technology - CAPG

2021.9 – 2023.12

B.S. - Software Engineering

Zhejiang Gongshang University - College of Computer Science and Technology

2017.9 – 2021.6

- First Prize, 2020 Zhejiang Provincial Undergraduate Mathematics Competition
- First Prize in the preliminary round, 2020 National Undergraduate Mathematics Competition

RESEARCH INTERESTS

3D Human Motion Synthesis

Generative Model

Video Generation / Editing